

CASE STUDY · TAKE-HOME EXERCISE

Senior AI Engineer

Building a Grounded Conversational Assistant
over Membership & Relationship Data

6 days

Window to complete & submit

~4 hrs

Reference pace — not a hard cutoff

10 + 5

Core & bonus prompts

7 files

Synthetic dataset

The Brief

What you're building, and why

You're building a small conversational assistant for a fictional membership organization. Internal team members should be able to ask it about accounts, people, events, activities, and source evidence - and trust what it tells them.

All data is 100% fictitious

Names, companies, events, notes, and relationships were invented for this exercise. Treat the files in data/ as your complete, closed knowledge base - no external lookups.

WHAT WE'RE EVALUATING

- ✓ **Engineering judgment**
- ✓ **Grounding & citations**
- ✓ **Handling of uncertainty**

We care far more about this than UI polish. A CLI or notebook is a completely valid deliverable.

What Your Assistant Must Do

Eight behaviors we'll be looking for across your answers

1

Answer only from provided files

No outside knowledge, no invented facts, dates, or figures.

2

Cite record & source IDs

Every claim should be traceable back to a specific row or source.

3

Separate public vs. internal sources

Never present an internal note as if it were public fact.

4

Flag weak evidence

Say when something is missing, stale, conflicting, or low-confidence.

5

Protect restricted contact details

Respect each person's contact_visibility setting.

6

Ask when a request is ambiguous

Clarify rather than guessing which person, account, or event.

7

Choose the right response mode

Answer directly, ask a follow-up, refuse, or escalate to a human.

8

Track session vs. dataset

Keep temporary notes distinct from the original CSVs.

The Dataset

Seven linked CSV files, described in full in data_dictionary.md

accounts.csv

18 rows

Company/account master data - sector, ownership, membership status

people.csv

41 rows

People linked to accounts - role, family status, contact visibility

events.csv

7 rows

Event records - city, date, type, status

event_attendance.csv

25 rows

RSVP & attendance status per person, per event

activities.csv

25 rows

CRM-style activity log - meeting notes, public updates, research

relationships.csv

4 rows

Possible relationship paths between people & accounts

sources.csv

40 rows

Source snippets - type, reliability, and visibility for every record

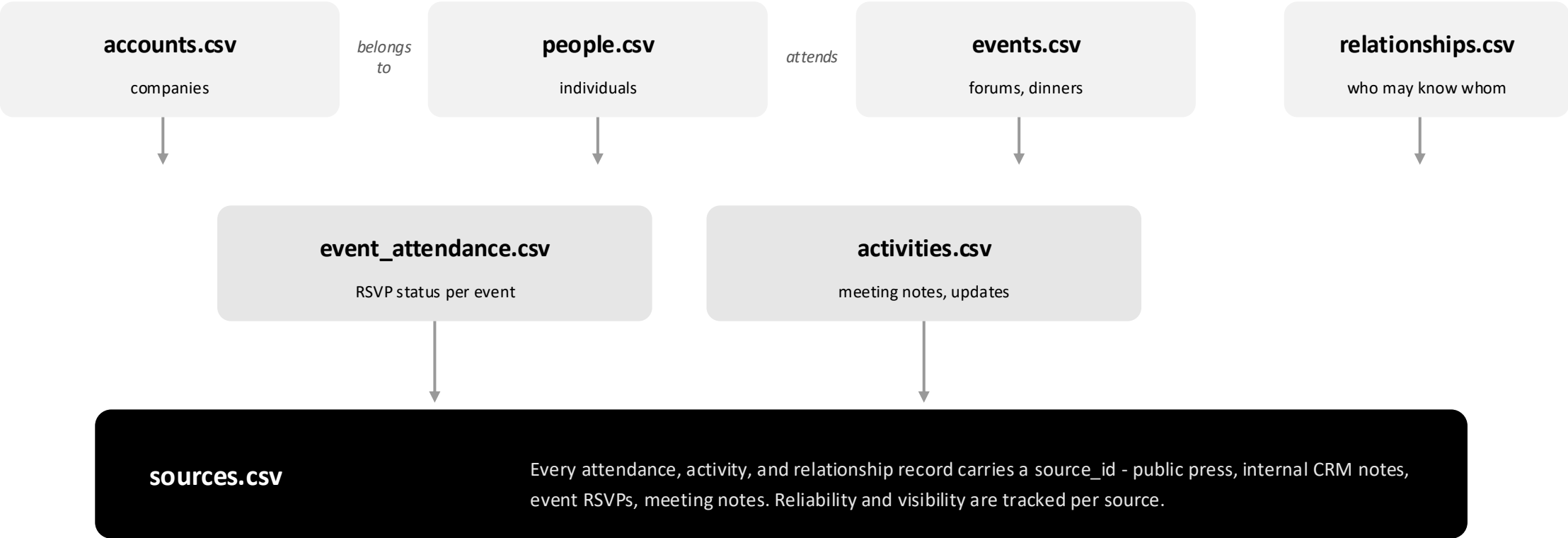
policies.md + data_dictionary.md

reference

Rules to follow, and the full schema for every field

How the Data Connects

Every operational record points back to people and accounts and cites a source



Full field-by-field detail lives in `data_dictionary.md` this is the shape, not the schema.

Policies You Must Follow

The full rules live in policies.md: This is the summary

Grounding

Never invent facts, dates, attendance, revenue, or relationships. Say when the dataset doesn't support an answer.

Conflicts

When records disagree, flag it and cite both sides. Prefer newer evidence, but don't silently drop the old.

Sensitive topics

Succession, ownership transition, and family/health matters need cited, reliability-labeled answers and a human-review flag for rumors or restricted notes.

Ambiguity

If a request could mean multiple people, accounts, or events, ask — or present the plausible options with citations.

CONTACT-DETAIL VISIBILITY

shareable_business

Business email & phone may be shown

email_only

Email may be shown, phone must not

restricted

Neither shown: say permission is required

internal_only

Not shown externally: summarize as internal-only

The Prompts

Full wording in prompts.md; bonus prompts are optional; answer as many as time allows

10 CORE PROMPTS

- 1 Member guests attending the Berlin Manufacturing Forum
- 2 Session-update Northstar Holdings with a voice note, then show recent activity
- 3 Accounts with recent succession/ownership activity in Asia
- 4 German family-owned industrials active in data centers
- 5 Prep call notes for an upcoming Valen Group call
- 6 Is Meridian Foods attending the Berlin dinner?
- 7 Has Cardso Precision been involved in Asian events?
- 8 Who potentially knows someone at Hansei Textiles?
- 9 What do we know about Priya Kapoor?
- 10 Give me Priya Kapoor's contact details

5 BONUS PROMPTS

Optional stretch questions — testing ambiguity, sensitivity, and unsupported claims.

- 1 Which families are preparing for a sale or ownership change?
- 2 Is Valen Group active in data centers?
- 3 Tell me everything about Daniel Weber's succession plans
- 4 Who should I contact about the Singapore dinner?
- 5 Is Northstar Holdings active in renewable packaging?

Deliverables Checklist

Seven short notes plus a runnable prototype - bullet points are fine



Runnable prototype

CLI, notebook, API, or lightweight web app — clearly executable



Context / memory note

How conversation context is carried across turns



Session-notes note

How temporary user notes are stored, cited, and kept separate



Evaluation plan

How you'd know the assistant is working well



Architecture note

A short explanation of your overall approach



Retrieval strategy note

How the assistant accesses and searches the data



Decision-logic note

When it answers, follows up, refuses, or escalates to a human



Privacy & limitations note

Governance considerations and known gaps

3–5 short bullet points per note is enough — we're reading for judgment, not documentation length.

Getting Started

Practical guidance before you open your editor

1

6 days to submit, ~4 hrs as your pace guide

The 4-hour figure is a reference for scoping your effort, not a hard cutoff. Take the time you need within the 6-day window.

2

Any stack works

Framework, no framework, any model provider. A strong solution can be simple.

3

data/ is your complete world

No external lookups. Treat the CSVs and policies.md as the whole knowledge base.

4

Session notes ≠ the dataset

Handle temporary, conversation-provided updates distinctly from the original files.

Questions about the exercise? Reach out directly to Andreea at any point. Asking is part of good judgment, not a penalty.